1.40 Trigonometry: sin(x), cos(x), tan(x) in Degrees

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sin/cos/tan of $30^\circ, 45^\circ, 60^\circ$

3

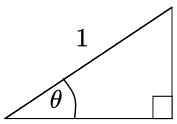
You will frequently encounter angles of 30°, 60°, 45° in geometric problems. Why?

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The Unit Circle and Trigonometry

4

For values of θ in the range $0 < \theta < 90^\circ$, you know that $\sin\theta$ and $\cos\theta$ are lengths on a right-angled triangle:



1.40 Trigonometry: sin(x), cos(x), tan(x) in Degrees

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Mini-Exercise

5

Use the unit circle to determine each value in the table, **using either "0", "+ve", "-ve", "1", "-1" or "undefined"**.
Recall that the point on the unit circle has coordinate $(\cos\theta, \sin\theta)$ and OP has gradient $\tan\theta$.

	$\cos\theta$	$\sin\theta$	$\tan\theta$
$\theta = 0$ 	1	0	0
$0 < \theta < 90^\circ$ 			
$\theta = 90^\circ$ 			
$90^\circ < \theta < 180^\circ$ 			

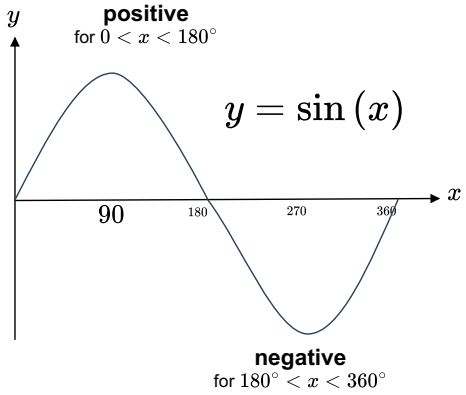
	$\cos\theta$	$\sin\theta$	$\tan\theta$
$\theta = 180^\circ$ 			
$180^\circ < \theta < 270^\circ$ 			
$\theta = 270^\circ$ 			
$270^\circ < \theta < 360^\circ$ 			

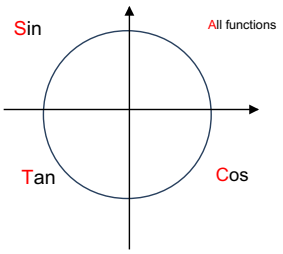
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The Unit Circle and Trigonometry

6

The unit circles explains the behaviour of these trigonometric graphs beyond 90°.
However, the easiest way to remember whether $\sin(x), \cos(x), \tan(x)$ are positive or negative is to just do a **very quick sketch (preferably mentally!)** of the corresponding graph.





Note: The textbook uses something called '*CAST diagrams*', which is an acronym for which quadrants the **C**osine, **A**ll, **S**ine and **T**angent are positive

1.40 Trigonometry: sin(x), cos(x), tan(x) in Degrees

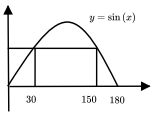
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A Few Trigonometric Angle Laws

7

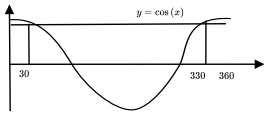
The following are all easily derivable using a quick sketch of a trigonometric graph, and are merely a convenience so you don't always have to draw out a graph every time.
You are highly encouraged to **memorise these** so that you can do exam questions faster.

1. $\sin(x) = \sin(180^\circ - x)$
   e.g. $\sin(150^\circ) = \sin(30^\circ)$

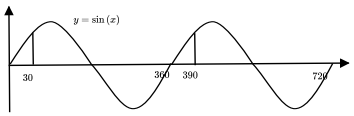


We saw this in the previous chapter when covering the 'ambiguous case' when using the sine rule.

2. $\cos(x) = \cos(360^\circ - x)$
   e.g. $\cos(330^\circ) = \cos(30^\circ)$



3. $\sin$ and $\cos$ repeat every 360° but $\tan$ every 180°
   e.g. $\sin(390^\circ) = \sin(30^\circ)$



4. $\sin(x) = \cos(90^\circ - x)$
   e.g. $\sin(50^\circ) = \cos(40^\circ)$

Remember from the previous chapter that "cosine" by definition is the sine of the "complementary" angle. This was/is never covered in the textbook but caught everyone by surprise when it came up in a C3 exam.

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Examples

8

Without a calculator, work out the value of each below.

$\tan(225^\circ) =$

$\tan(210^\circ) =$

$\sin(150^\circ) =$

$\cos(300^\circ) =$

$\sin(-45^\circ) =$

$\cos(750^\circ) =$

$\cos(120^\circ) =$

Use the 'laws' where you can, but otherwise just draw out a quick sketch of the graph.

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- $\sin, \cos$ repeat every 360° but $\tan$ every 180°

1.40 Trigonometry: sin(x), cos(x), tan(x) in Degrees

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Test Your Understanding

9

Without a calculator, work out the value of each below.

$$\cos(315^\circ) =$$

$$\sin(420^\circ) =$$

$$\tan(-120^\circ) =$$

$$\tan(-45^\circ) =$$

$$\sin(135^\circ) =$$

- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- $\sin, \cos$ repeat every 360° but $\tan$ every 180°

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Exercise 10B – Y1 Page 209

10

1 Express the following as trigonometric ratios of either 30°, 45° or 60°, and hence find their exact values.\quad
a $\sin 135^\circ$\quad**b** $\sin(-60^\circ)$\quad**c** $\sin 330^\circ$\quad**d** $\sin 420^\circ$\quad**e** $\sin(-300^\circ)$\quad**f** $\cos 120^\circ$\quad**g** $\cos 300^\circ$\quad**h** $\cos 225^\circ$\quad**i** $\cos(-210^\circ)$\quad**j** $\cos 495^\circ$\quad**k** $\tan 135^\circ$\quad**l** $\tan(-225^\circ)$\quad**m** $\tan 210^\circ$\quad**n** $\tan 300^\circ$\quad**o** $\tan(-120^\circ)$.

t

t

1.40 Trigonometry: sin(x), cos(x), tan(x) in Degrees

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Exercise 10A/B – Y1 Page 207, 209

11

Challenge

The diagram shows an isosceles right-angled triangle ABC.

$AE = DE = 1$ unit. Angle $ACD = 30°$.

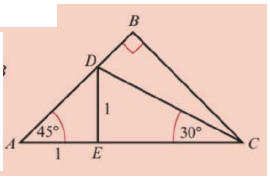
a Calculate the exact lengths of

i CE \quad ii DC \quad iii BC \quad iv DB

b State the size of angle BCD.

c Hence find exact values for

i $\sin 15°$ \quad ii $\cos 15°$



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Solving Trigonometric Equations

19

Remember those trigonometric angle laws (on the right) earlier this chapter? They're about to become very useful!

Reminder of 'trig laws':
- $\sin(x) = \sin(180 - x)$
- $\cos(x) = \cos(360 - x)$
- sin, cos repeat every $360°$ but tan every $180°$

Solve $\sin\theta = \frac{1}{2}$ in the interval $0 \leq \theta \leq 360°$.

Solve $5\tan\theta = 10$ in the interval $-180° \leq \theta < 180°$

1.40 Trigonometry: sin(x), cos(x), tan(x) in Degrees

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Slightly Harder Ones…

20

Solve $\sin\theta = -\frac{1}{2}$ in the interval $0 \leq \theta \leq 360^\circ$.

In general you should have 2 solutions per 360° (except when at a peak or trough of the trig graph)

Note that we've had to use a second law, i.e. that $\sin$ repeats every 360°.

Solve $\sin\theta = \sqrt{3}\cos\theta$ in the interval $0 \leq \theta \leq 360^\circ$.

Hint: The problem here is that we have two different trig functions. Is there anything we can divide both sides by so we only have one trig function?

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Test Your Understanding

21

Solve $2\cos\theta = \sqrt{3}$ in the interval $0 \leq \theta \leq 360^\circ$.

Solve $\sqrt{3}\sin\theta = \cos\theta$ in the interval $-180^\circ \leq \theta \leq 180^\circ$.

1.40 Trigonometry: sin(x), cos(x), tan(x) in Degrees

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Exercise 10D – Y1 215-216

22

7 A teacher asks two students to solve the equation $2\cos x = 3\sin x$ for $-180^\circ \leq x \leq 180^\circ$. The attempts are shown:
Student A: $\tan x = \frac{3}{2}$, $x = 56.3^\circ$ or $x = -123.7^\circ$.
Student B: $4\cos^2 x = 9\sin^2 x$, $4(1 - \sin^2 x) = 9\sin^2 x$, $4 = 13\sin^2 x$, $\sin x = \pm\sqrt{\frac{4}{13}}$, $x = \pm 33.7^\circ$ or $x = \pm 146.3^\circ$.
a Identify the mistake made by Student A. (1 mark)
b Identify the mistake made by Student B and explain the effect it has on their solution. (2 marks)
c Write down the correct answers to the question. (1 mark)

9 Find all the values of θ, to 1 decimal place, in the interval $0 < \theta < 360^\circ$ for which $\tan^2\theta = 9$

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Exercise 10D – Y1 215-216

23

10 a Show that $4\sin^2 x - 3\cos^2 x = 2$ can be written as $7\sin^2 x = 5$.
b Hence solve, for $0 \leq x \leq 360^\circ$, the equation $4\sin^2 x - 3\cos^2 x = 2$. Give your answers to 1 decimal place.

11 a Show that the equation $2\sin^2 x + 5\cos^2 x = 1$ can be written as $3\sin^2 x = 4$.
b Use your result in part a to explain why the equation $2\sin^2 x + 5\cos^2 x = 1$ has no solutions.

1.40 Trigonometry: sin(x), cos(x), tan(x) in Degrees

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Harder Equations

24

Harder questions replace the angle θ with a linear expression.

Solve $\cos 3x = -\frac{1}{2}$ in the interval $0 \leq x \leq 360^\circ$.

STEP 1: Adjust the range of values for θ to match the expression inside the cos.

STEP 2: Immediately after applying an inverse trig function (and BEFORE dividing by 3!), find all solutions up to the end of the interval.

STEP 3: Then do final manipulation to each value.

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Further Examples

25

Solve $\sin(2x + 30^\circ) = \frac{1}{\sqrt{2}}$ in the interval $0 \leq x \leq 360^\circ$.

To get from x to $2x + 30^\circ$ we double and add 30°. So do the same to the upper and lower bound!

Solve $\sin x = 2\cos x$ in the interval $0 \leq x < 300^\circ$

By dividing both sides by $\cos x$, the $\sin x$ becomes $\tan x$ and the $\cos x$ disappears, leaving a trig equation helpfully only in terms of one trig function.

1.40 Trigonometry: sin(x), cos(x), tan(x) in Degrees

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Test Your Understanding

26

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Solve, for $0 \leq x < 180^\circ$,
$\cos(3x - 10^\circ) = -0.4$,
giving your answers to 1 decimal place. You should show each step in your working. (7)

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Exercise 10E – Y1 p219

27

4 Solve for $0 \leq x \leq 180^\circ$ the equations:
a $\sin(x + 20^\circ) = \frac{1}{2}$.
b $\cos 2x = -0.8$, giving your answers to 1 decimal place.

5 a Sketch for $0 \leq x \leq 360^\circ$ the graph of $y = \sin(x + 60^\circ)$.
b Write down the exact coordinates of the points where the graph meets the coordinate axes.
c Solve, for $0 \leq x \leq 360^\circ$, the equation $\sin(x + 60^\circ) = 0.55$, giving your answers to 1 decimal place.

1.40 Trigonometry: sin(x), cos(x), tan(x) in Degrees

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Exercise 10E – Y1 p219

28

6 a Given that $4\sin x = 3\cos x$, write down the value of $\tan x$.
b Solve, for $0 \leq \theta \leq 360^\circ$, $4\sin 2\theta = 3\cos 2\theta$ giving your answers to 1 decimal place.

7 The equation $\tan kx = -\frac{1}{\sqrt{3}}$, where k is a constant and $k > 0$, has a solution at $x = 60^\circ$.
a Find a possible value of k.
b State, with justification, whether this is the only such possible value of k.